ST422 Report: Central England Seasonal Temperature Change

Team 3: Yueran Wang 2201370

2026-04-17

Contents

0.1 Contribution declaration . . . 3

1 Client report **3**
1.1 Executive summary . . . 3
1.2 Client brief and scope . . . 4
1.3 Data and context . . . 4
1.4 Findings . . . 4
1.4.1 Recent decades are warmer than a generation ago . . . 4
1.4.2 Warming has been sustained across the whole historical record . . . 5
1.4.3 The pace of warming strengthened from the mid-1970s . . . 6
1.4.4 A monitoring framework is ready to use . . . 6
1.5 Limitations, Recommendations and Future work . . . 7

2 Technical Report **8**
2.1 Technical Abstract . . . 8
2.2 Client Decision Context, Scope and Estimands . . . 8
2.3 Data and Pre-processing . . . 8
2.3.1 Data Provenance, Structure, and Analytical Scope . . . 8
2.3.2 Inclusion/exclusion Rules and Known Data Limitations . . . 8
2.3.3 Pre-processing and Feature Construction . . . 8
2.4 Exploratory Analysis and Descriptive Checks . . . 9
2.4.1 EDA Objectives . . . 9
2.4.2 Annual Records and Long-Run Context . . . 9
2.4.3 Rolling Warming Rates Through Time . . . 9
2.5 Methodology of Modelling . . . 10
2.5.1 Model choice & Assumptions . . . 10
2.5.2 GLS AR(1): Baseline Shift Model . . . 10
2.5.3 GLS AR(1): Long-Run Trend Model . . . 11
2.5.4 Serial Dependence in Baseline and Long-Run Models . . . 12
2.5.5 Model Comparison and Comparison Criteria . . . 13
2.6 Validation and Robustness . . . 14
2.6.1 Residual Diagnostics . . . 14
2.6.2 Robustness Checks . . . 14
2.7 Monitoring and Material-Change Thresholds . . . 16
2.8 Limitations, Validity, and Alternatives . . . 16
2.8.1 Threats to Validity and Limitations . . . 16
2.8.2 Failure Mode That Would Change the Conclusion . . . 17
2.8.3 Alternative Approaches and Conditions for Preference . . . 17
2.9 Traceability and Reproducibility . . . 17
2.9.1 Claim-to-Evidence Map for Decision 1, 2, and 3 . . . 17
2.9.2 Simplifications Planned for the Client-Facing Report . . . 17
2.9.3 Repository Structure, Run Sequence, Environment, and Determinism . . . 17
2.9.4 Output Inventory and Remaining Reproducibility Risks . . . 18
2.10 Appendix: Individual reflection . . . 19
2.10.1 Key judgement calls you made . . . 19
2.10.2 Trade-offs . . . 19
2.10.3 What you learned and what you would do differently . . . 19
2.11 Appendix: Evidence of group contact and professional working . . . 19
2.12 Appendix: Management plan . . . 27

0.1 Contribution declaration

Summary of my main contributions

1. I produced the core EDA and modelling visualisations used to support both the technical interpretation and the client-facing narrative. The main evidence is in commits `ae12467`, `7811600`, `9d45e38`, and PR `#28`.
2. I led the data analysis and modelling work for the project, including the baseline shift and long run trend analysis, the model comparison workflow, and the monitoring threshold framework. The main code evidence is in commits `027186d`, `5f33013`, and PR `#28`.
3. I carried out the robustness and diagnostics work used to test whether the headline findings were stable to alternative specifications and unusual seasons, and I integrated those results into the report. The main code evidence is in commits `5f33013`, `9d45e38`, and PR `#28`.
4. I managed the later versioned redevelopment of the `src` workflow, moving the project from earlier script based outputs to a modular structure organised into `models`, `plots`, `tables`, and `utils`, and then aligning the project files with that structure. The main evidence is in PR `#24`, commit `f0d1b71`, and the follow-up commits `5f33013`, `9d45e38`, `8104b71`, `1fcfee8`, and `5d25022`.

Items I reviewed or quality-assured

1. I reviewed the integration of plots, tables, utilities, and report outputs during the later consolidation of the repository structure. The main evidence is in commits `8104b71`, `1fcfee8`, and `5d25022`.
2. I reviewed the consistency between the analytical outputs and the written report during the week 9 reporting update. The main evidence is in PR `#12`, commit `ec409c9`, and PR `#28`.

Evidence pointers

- Audit summary: 20 commits and 8 pull requests under `Yueran-Wang-Warwick`
- Key pull requests: `#12 Yueran week9`, `#23 Update attendees' status for March 11 meeting`, `#28 Update for eda, modelling, and robustness check`
- Key issues: `#5 Progress Update: EDA visualisation and baseline analysis`, `#8 Progress Update for week 7 by Yueran`
- Key commits: `027186d`, `ae12467`, `7811600`, `f0d1b71`, `5f33013`, `9d45e38`, `8104b71`, `1fcfee8`, `5d25022`
- Version management sequence for `src`: early visualisation and modelling work in `ae12467`, `7811600`, and `027186d`, restructuring activity in PR `#24` and commit `f0d1b71`, then modular redevelopment through `5f33013`, `9d45e38`, `8104b71`, and `1fcfee8`, with project level alignment in `5d25022`

1 Client report

1.1 Executive summary

The client question: The brief asked for an evidence-based view of how Central England temperatures have changed across the four seasons, and what those changes mean for internal planning. Three practical questions drove the work, covering whether recent decades are measurably warmer than an earlier generation, which seasons have changed the most, and what monitoring thresholds to use in seasonal briefings.

Headline findings with Evidence Anchors:

- Spring temperatures averaged roughly 1.1 degrees Celsius higher in the 1991 to 2020 period than in 1961 to 1990, the largest shift of the four seasons. Summer and autumn also show clear increases. Winter points in the same direction with a smaller and less certain shift (see Figure 1).
- Over the full historical record from 1879 to 2024, every season has been warming at a sustained long-run rate. All four seasons fall in the range of about 0.09 to 0.13 degrees Celsius per decade, with autumn warming fastest and summer warming slowest (see Figure 2).
- The pace of warming strengthened after around 1975. Warming rates estimated over the most recent decades are consistently higher than those calculated over the full historical record in every season (see Figure 3).
- A seasonal monitoring framework has been defined, assigning any incoming season a status of Reference, Watch, Action, or Escalation based on how far that season sits from the 1991 to 2020 reference distribution (Figure 4).

Recommendation: We recommend adopting the 1991-2020 period as the stand-alone reference baseline for seasonal planning and using the established normal seasonal range shown in Figure 4 as the operating band for routine monitoring. This gives internal briefings a clear, ready-to-use decision rule for when an incoming season warrants attention and when it does not. Any season falling noticeably outside that range should be flagged for follow-up. Winter conclusions

should be treated with marginally more caution than the other three seasons, as its estimated shift is less precisely determined.

Key uncertainties and Limitations: The broad direction of change is clear, but the exact size of that change is less certain for unusually extreme seasons. The statistical checks suggest that the main pattern is reliable, whilst also indicating that the uncertainty around the most exceptional seasons may be slightly larger than the reported ranges suggest. The evidence is based on seasonal averages for Central England as a whole, so a season that looks typical in the overall data could still contain shorter periods of unusually hot or cold weather. The monitoring thresholds should also be read as practical decision rules for internal briefings, which means that different threshold choices would lead to alerts being triggered at different rates.

1.2 Client brief and scope

The brief requested a clear, evidence-based assessment of seasonal temperature change in Central England using the Met Office HadCET records. The practical goal was to support three linked planning decisions, covering whether a sustained shift in seasonal temperatures has taken place over recent decades, which seasons have changed the most and whether the change looks more like a gradual trend or a distinct step shift, and what metrics and thresholds to use for internal monitoring and communication.

What this analysis covers and not cover:

The analysis uses HadCET seasonal temperature records from 1879 to 2024, covering mean, minimum, and maximum temperatures across winter, spring, summer, and autumn. The primary baseline comparison is between a recent 30-year reference period (1991 to 2020) and an earlier historical period (1961 to 1990). Long-run warming rates are estimated over the full historical window. A shorter 10-year window covering 2015 to 2024 is also used as a near-term signal to detect whether the most recent decade sits above the 30-year reference distribution.

This report does not analyse sub-regional variation within Central England. It works with seasonal-level temperature summaries, so short-duration events such as individual heatwaves or cold spells at a daily or weekly scale are not captured. It also does not make projections about future temperature levels.

1.3 Data and context

The analysis draws on three official seasonal temperature series from Met Office (2025), *Seasonal Mean, Minimum and Maximum Central England Temperature (HadCET) series* v2.1.0.0, NERC EDS Centre for Environmental Data Analysis, doi:10.5285/ca43505702fa4eeeba4b65f1ce2c1e6a, covering seasonal mean, minimum, and maximum temperatures for the Central England region. Each series provides one value per season per year across a very long historical span. The mean series extends back to 1659, whilst the minimum and maximum series each begin in 1878. Because comparing all three temperature types requires overlapping records, all quantitative analysis in this report uses the shared period from 1879 to 2024.

The main data quality issue is structural rather than random. The first winter observation is unavailable because a winter value requires the previous December. The series are also regional seasonal summaries, so they smooth over short extreme episodes and do not show variation within Central England. Any measurement uncertainty in such a long historical record is more likely to affect precision than direction, which means the results are most useful for judging broad seasonal change rather than fine local detail.

1.4 Findings

1.4.1 Recent decades are warmer than a generation ago

Recent decades were warmer than a generation ago in every season and across all three temperature measures. Spring shows the largest increase, with its mean temperature rising by roughly 1.1 degrees Celsius. Summer and autumn also show clear upward shifts across mean, minimum, and maximum temperature. Winter is positive across all three temperature types, whilst its magnitude is harder to pin down with precision.

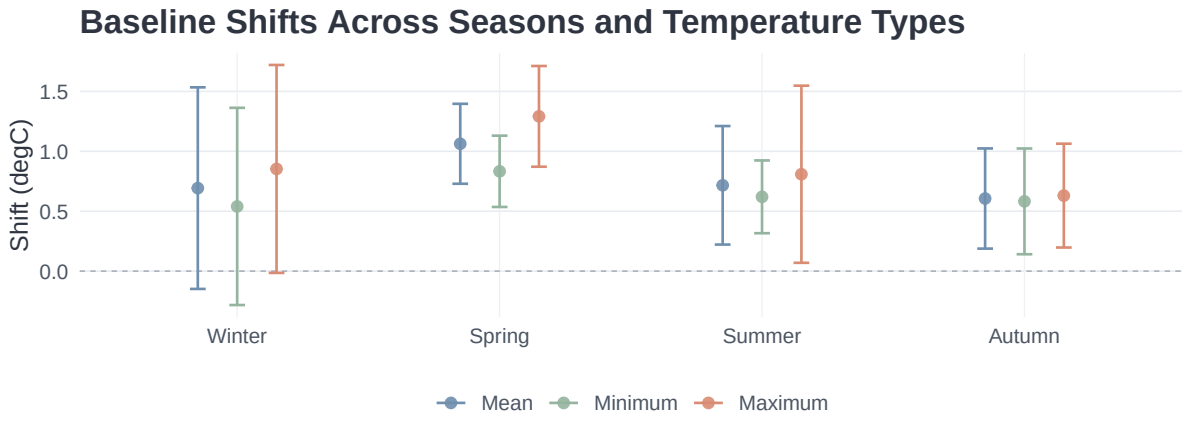

Figure 1: GLS AR(1) baseline shift estimates with 95% intervals by season and temperature type.

Figure 2 has four rows, one for each season, with three dots per row showing the estimated temperature increase for mean, minimum, and maximum. The horizontal axis shows the size of that increase in degrees Celsius, and the horizontal bars through each dot show the range of plausible values around the estimate. Every dot sits to the right of zero, confirming that all twelve season-temperature combinations were warmer in 1991 to 2020 than in 1961 to 1990. Spring's dots sit furthest right, and the bars for spring, summer, and autumn lie entirely in positive territory, whilst for winter the bars extend back close to zero, which is why the winter finding carries more caution than the other three seasons. The practical implication is that planning assumptions based on the older 1961 to 1990 baseline now understate current seasonal conditions, especially in spring.

1.4.2 Warming has been sustained across the whole historical record

Warming has been sustained across the whole historical record rather than appearing only in the most recent decades. Estimated long-run warming rates range from about 0.09 degrees Celsius per decade for summer to about 0.13 degrees per decade for autumn. Accumulated over 145 years, those rates correspond to total seasonal warming of roughly 1.3 to 1.9 degrees Celsius. All twelve combinations of season and temperature type give a positive long-run rate, with no exceptions.

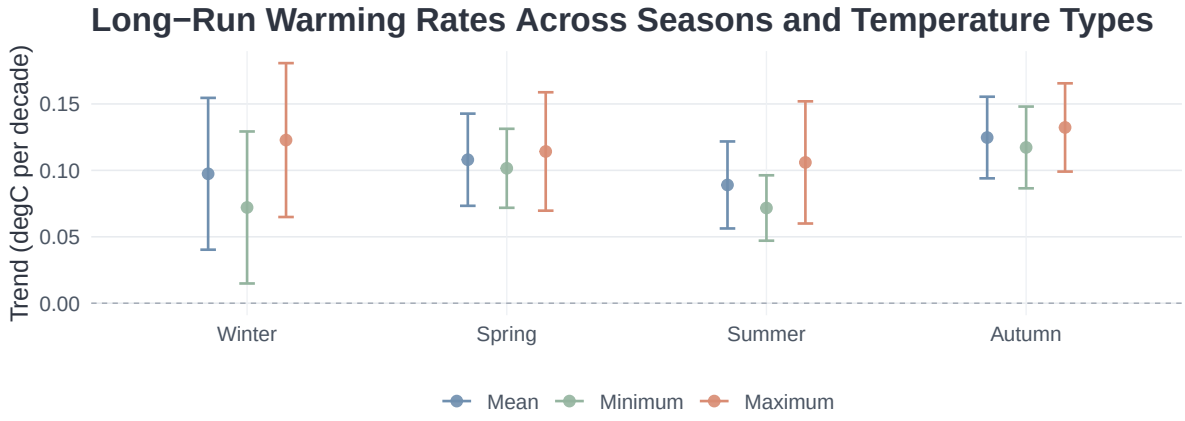

Figure 2: GLS AR(1) long-run warming rates with 95% intervals by season and temperature type.

Figure 3 follows the same layout as Figure 2, but the horizontal axis now shows the average rate of warming per decade over the full 1879 to 2024 period. The key difference from Figure 2 is that every bar in Figure 3 sits entirely to the right of zero, including winter, which in Figure 2 was the one uncertain case. Autumn tends to sit furthest right and summer furthest left, reflecting the seasonal ranking in the long-run rates. The clear separation of all bars from zero confirms that the warming pattern is a genuine long-run feature and is not simply a consequence of the particular 30-year windows chosen for comparison. The practical implication is that the client should treat the observed shift as a persistent change in background conditions, not as a short-lived fluctuation that can be ignored.

1.4.3 The pace of warming strengthened from the mid-1970s

Warming rates moved into clearly positive territory from around 1975 onward across all four seasons and have remained there. Around 2010, rates in several seasons eased back slightly from their peak, but stayed positive.

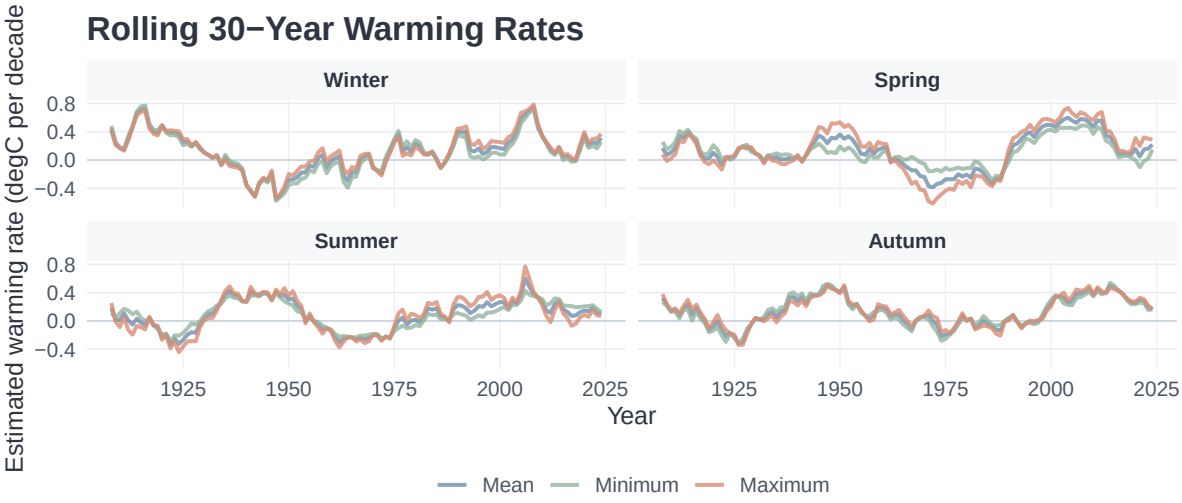

Figure 3: Rolling 30-year warming rates by season.

Figure 4 shows one panel per season, with time on the horizontal axis and the estimated warming rate within each trailing 30-year period on the vertical axis. Positive values mean that period was warming, negative values reflect a temporary cooling phase within the window. The lines for all three temperature types cross into positive territory and stay there from around 1975 to 1990 onward. Winter shows the deepest dip into negative rates around 1950, which explains why its long-run average in Figure 3 is lower than autumn's despite both finishing clearly positive by the present. The slight easing of rates after about 2010 is why the 10-year reference window (2015 to 2024) is useful as a leading check on whether current seasons are still tracking at, or above, the primary 30-year reference level. The practical implication is that routine monitoring should not rely on a single long baseline alone, because a shorter recent window provides an earlier signal when conditions are continuing to shift.

1.4.4 A monitoring framework is ready to use

A practical monitoring framework is ready to use for internal seasonal briefings. Figure 4 shows the usual recent range for each season and temperature type under both the main 30 year reference period and the more recent 10 year period. This range excludes the coolest 10 per cent and the warmest 10 per cent of recent seasons, so values outside it can be treated as unusually cool or unusually warm in relation to the recent baseline. That makes it easier to judge whether the newer period is already sitting above the main planning baseline.

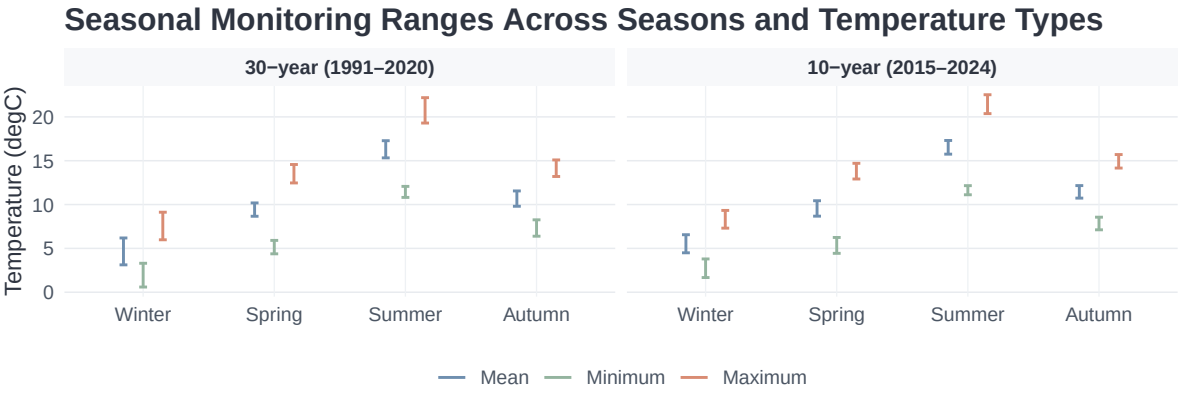

Figure 4: Seasonal monitoring ranges from the 30-year and 10-year reference windows

Any incoming season can be assigned one of four status levels by comparing its observed value with the usual recent range shown in Figure 4. A season that falls within that range is **Reference** status, which means conditions remain within the recent norm. A single season outside that range is **Watch** status and should be flagged for follow up in the next briefing. Repeated breaches, or a particularly large single breach, move the season to **Action** status and call for an escalation note. If at least two of the three temperature measures reach Action status in the same season, the framework moves to **Escalation** and triggers a wider planning discussion. The practical implication is direct, the client can use this framework now to decide when a season should simply be noted, when it should be monitored more closely, and when it should trigger a planning response.

1.5 Limitations, Recommendations and Future work

The overall direction of change is clear, but some parts of the picture are more certain than others. Winter shows a relatively weaker recent baseline signal than spring, summer, and autumn, so winter specific decisions should be made with slightly more caution. The statistical checks also suggest that the main pattern is reliable, whilst indicating that uncertainty may be somewhat larger for the most unusual seasons than the reported ranges alone imply. In addition, the evidence is based on seasonal averages for Central England as a whole, so it does not show shorter extreme episodes within a season or differences between smaller local areas.

What the additional checks show: Additional checks were used to test whether the main findings depended too heavily on one modelling choice or a small number of unusual years. The results were rechecked using alternative model forms and by removing a small set of especially extreme seasons. These checks did not change the main conclusion. The evidence still supports warming across all four seasons, and the case for updating planning benchmarks remains unchanged. The main effect of these checks was on precision rather than direction.

What would reduce uncertainty further: Uncertainty would be reduced by using data with finer time and space detail. Monthly or local area data would show whether changes seen in the seasonal regional averages are also being driven by shorter extreme episodes or by stronger change in particular parts of Central England.

Three recommendations follow from the analysis. First, the client should adopt the 1991 to 2020 period as the standard reference baseline for internal seasonal planning, because the older 1961 to 1990 baseline no longer reflects current conditions. Second, the client should use the monitoring framework in Figure 4 in routine seasonal briefings, so that each season can be judged against a clear recent reference range and escalated when needed. Third, spring and autumn should receive the highest planning priority, because spring shows the largest recent shift relative to the older baseline, whilst autumn has the strongest long run warming rate. These actions should be applied consistently across internal reporting and seasonal review so that briefing messages remain aligned. The framework should be used as an early warning tool for briefing and review, and winter results should be interpreted with slightly more caution than those for the other seasons.

The next step in the short term is to apply the framework to the most recently completed season and assign a status level for each temperature measure. The next step over the longer term is to extend the analysis to data with finer time or spatial detail. Monthly data would help identify whether seasonal change is being driven by shorter extreme episodes, whilst more local data would show whether some parts of Central England are changing faster than others.

2 Technical Report

2.1 Technical Abstract

This report addresses the client question of whether Central England seasonal temperatures have shifted in recent decades, which seasons show the strongest change, and which monitoring thresholds should guide internal planning. Using Met Office HadCET seasonal mean, minimum, and maximum series (v2.1.0.0), the analysis combines EDA with season-metric specific GLS models under AR(1) errors to estimate baseline shifts (1991 to 2020 vs 1961 to 1990) and long-run trends (1879 to 2024). The headline conclusion is consistent across methods. Recent shifts are positive across all season-metric combinations, spring shows the largest baseline increases, winter is the least precise baseline case, and long-run trends are positive in all 12 combinations with post-1975 acceleration evident in sensitivity checks. The main limitations are simplified dependence modelling, visible winter-tail residual deviation, heuristic monitoring thresholds, and aggregation to seasonal regional series. These were handled by comparing OLS, AR(1), and AR(2) specifications, checking residual diagnostics directly, running segmented-trend and influential-year robustness checks, and presenting threshold rules as operational screening triggers rather than fixed optimal policy cut-offs.

2.2 Client Decision Context, Scope and Estimands

The client requires decision-ready evidence on three linked questions, whether recent seasonal temperatures have shifted relative to earlier baselines, which seasons show the strongest changes and in what form, and which monitoring metrics and thresholds should guide internal planning. The analysis targets the HadCET Central England seasonal series (Midlands representation), with year as the unit of analysis and seasonal mean, minimum, and maximum temperature (degC) as primary outcomes. For cross-metric comparability, the core window is 1879 to 2024. Two primary estimands are used, seasonal baseline shift between 1961 to 1990 and 1991 to 2020, and seasonal long-run trend in degC per decade over 1879 to 2024. Decision support is judged by coherence in direction, seasonal ranking, and uncertainty across descriptive and model-based evidence. Monitoring thresholds are operationalised in Section 2.7 using season-metric reference distributions.

2.3 Data and Pre-processing

2.3.1 Data Provenance, Structure, and Analytical Scope

This analysis uses three archived HadCET seasonal totals files in this repository, corresponding to Met Office (2025), *Seasonal Mean, Minimum and Maximum Central England Temperature (HadCET) series* v2.1.0.0, NERC EDS Centre for Environmental Data Analysis, doi:10.5285/ca43505702fa4eeeba4b65f1ce2c1e6a. Each file provides year-indexed seasonal temperatures (winter, spring, summer, autumn, annual) in degC. In cleaned outputs, mean covers 1659 to 2024 (366 rows), whilst minimum and maximum each cover 1878 to 2024 (147 rows).

2.3.2 Inclusion/exclusion Rules and Known Data Limitations

Included data are the three HadCET seasonal totals series (mean, minimum, maximum). Seasonal ranked files are excluded because they are not the underlying time-series inputs for modelling. No external joins, covariates, or spatial merges are used. Known limitations are structural. The series is a Central England regional aggregate and does not resolve sub-regional Midlands variation. Seasonal values aggregate three-month periods, so within-season short-duration extremes are not directly represented. Coverage differs by metric (mean from 1659, minimum and maximum from 1878), and the first winter entry in each series is structurally missing because previous-December data are unavailable for the first recorded year.

2.3.3 Pre-processing and Feature Construction

Pre-processing removed non-data header and footer lines, retained the six analytical fields `year`, `winter`, `spring`, `summer`, `autumn`, and `annual`, converted `year` to integer, converted seasonal fields to numeric, and exported cleaned series for modelling. Missingness is structural rather than random. The only missing values are the first winter entry in each series, caused by the winter definition requiring previous-December data. No imputation was applied. Potentially influential observations were flagged using two rules within each season-metric series. The first rule used robust MAD z-scores with `|z| > 3.5`. The second rule used IQR fences with lower bound `Q1 - 1.5*IQR` and upper bound `Q3 + 1.5*IQR`. This flagged 30 observations, including winter 1963, summer 1976, and autumn 2006, 2022, and 2023. Flagged values were retained and evaluated through sensitivity checks in Section 2.6.2. After cleaning, the mean series contains 366 yearly records from 1659 to 2024, and the minimum and maximum series each contain 147 yearly records from 1878 to 2024. Cross-metric analysis uses the overlapping 1879 to 2024 window.

2.4 Exploratory Analysis and Descriptive Checks

2.4.1 EDA Objectives

The exploratory work addressed two main questions before any formal analysis was applied. The starting point was to examine the raw annual records alongside the long-run rolling averages and the two baseline windows, so that the direction of change and the context of the comparison could be assessed together in a single view. Once the overall direction was established, the next question was whether the pace of warming had remained stable across the record or had intensified in more recent decades.

2.4.2 Annual Records and Long-Run Context

Figure 5 plots the raw annual temperature records from 1879 to 2024 for each season, with rolling 30-year mean lines overlaid for each metric and the two comparison windows shaded in the background.

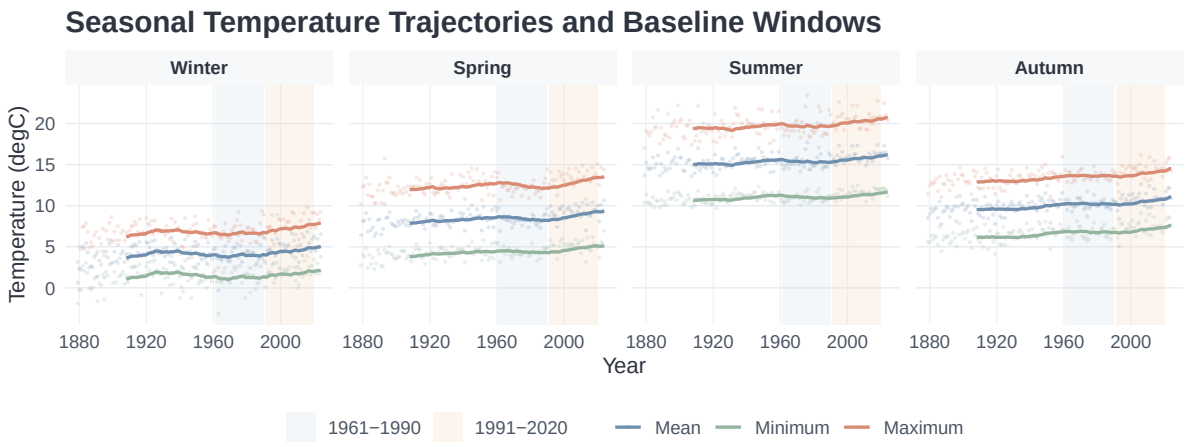

Figure 5: Annual HadCET seasonal temperatures by season, with rolling 30-year means.

The four panels correspond to the four seasons, each showing faint annual data points alongside three rolling 30-year mean lines for mean, minimum, and maximum temperature distinguished by colour, with the two baseline windows highlighted by shaded bands. The rolling means rise in every panel, with the rate of increase steepening in the later decades, and both shaded windows sit on the upward portion of each curve rather than at a stationary level. The consistent rise across all panels before, during, and after both windows establishes that the baseline shift is part of a sustained long-run process.

2.4.3 Rolling Warming Rates Through Time

Figure 6 below tracks the estimated warming rate within each trailing 30-year window. This turns the long-run trajectories in Figure 5 into a time-varying rate measure, so the record can be assessed not only for direction but also for whether the pace of warming has remained stable or intensified. It's clear that the pace of warming has not been constant over time. Around 1950, winter is the only season with a clearly negative rolling warming rate, indicating a temporary cooling phase within the 30-year window, whereas spring, summer, and autumn are already warming and follow broadly similar paths across most of the record.

Thereafter, all four seasons move into positive territory after about 1975, and negative rates become rare after 2000. At the same time, the three temperature metrics do not move equally closely together across seasons. Autumn shows the greatest overlap between mean, minimum, and maximum temperature, whilst spring shows the least. Winter and summer are more tightly grouped before about 1950, but separate more clearly afterwards. Therefore, these patterns indicate that temperatures have not only risen in recent decades, but that the rate of increase has also strengthened over time.

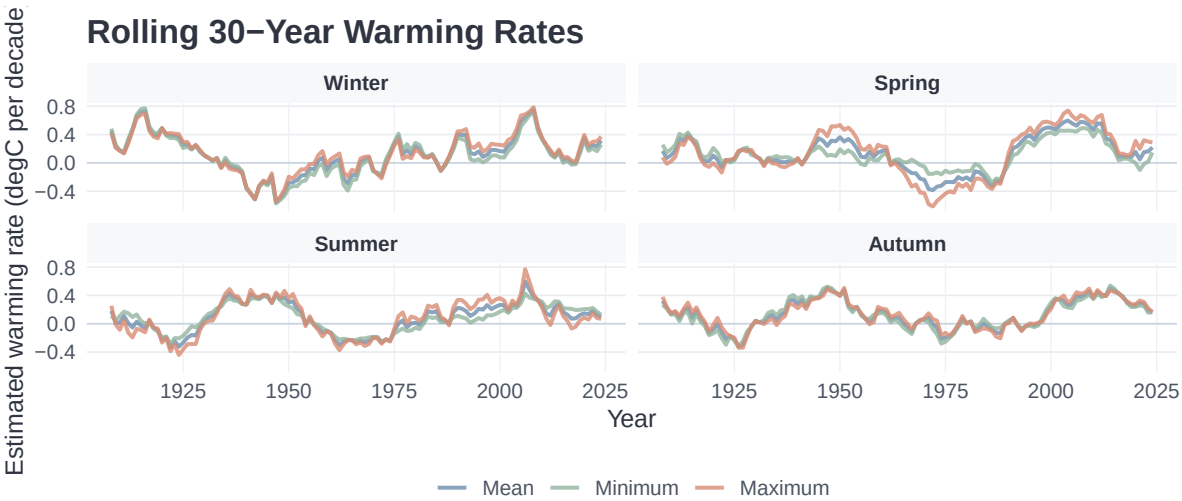

Figure 6: Rolling 30-year warming rates by season.

2.5 Methodology of Modelling

2.5.1 Model choice & Assumptions

The main models are fitted using GLS with an AR(1) error structure. This choice allows for year-to-year dependence in seasonal temperature series while remaining simple, interpretable, and directly comparable across seasons and temperature metrics.

2.5.2 GLS AR(1): Baseline Shift Model

The shift model estimates **how much warmer the 1991 to 2020 period was compared with 1961 to 1990 for a given season and temperature type**. For each of the 12 season-metric combinations, the model is fitted separately to 60 observations, comprising 30 annual values from the historical window and 30 from the recent window. The formal specification is

$$\text{temp}_t = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{recent}_t] + \varepsilon_t, \qquad \varepsilon_t = \rho\,\varepsilon_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

Each symbol has a direct interpretation:

- temp_t is the observed seasonal temperature in year t, the raw number in the dataset.
- β_0 is the mean temperature in the historical window, the **baseline level** against which the recent period is compared.
- The indicator $\mathbf{1}[\text{recent}_t]$ acts as a switch, equalling one for years 1991 to 2020 and zero for years 1961 to 1990.
- β_1 is therefore the primary estimand, representing **how much higher the recent window mean is compared with the historical window mean**, in degrees Celsius.
- The residual ε_t is the difference between that year's observed temperature and the average level for its window, and is modelled as an AR(1) process rather than assumed to be independent across years.
- The parameter ρ measures how much this year's "above or below average" carries over into next year. If ρ is positive, a year that is warmer than its window average makes the next year more likely to be warmer than its window average too, so adjacent observations are dependent rather than independent.
- η_t is the white-noise innovation remaining after removing that dependence, assumed to follow a normal distribution with mean zero and variance σ^2.

The four unknowns β_0, β_1, ρ, and σ^2 are estimated jointly using REML in R. The model is fitted 12 times in total, once for each combination of four seasons and three temperature types (mean, minimum, and maximum). Each fit uses an entirely separate set of 60 observations and produces its own estimates of β_1, ρ, and the associated confidence interval. Table 1 and Figure 7 collect all 12 results.

Table 1: GLS AR(1) baseline shift estimates (degC) with 95% CIs.

Season	β_1: Mean	β_1: Min	β_1: Max
Winter	+0.692 [-0.149, +1.534]	+0.540 [-0.283, +1.363]	+0.853 [-0.015, +1.721]
Spring	+1.063 [+0.729, +1.396]	+0.833 [+0.535, +1.130]	+1.292 [+0.871, +1.712]
Summer	+0.716 [+0.221, +1.211]	+0.620 [+0.316, +0.924]	+0.808 [+0.069, +1.548]
Autumn	+0.606 [+0.188, +1.024]	+0.582 [+0.141, +1.023]	+0.630 [+0.197, +1.063]

Summer minimum in Table 1 is used here as a worked example to illustrate how the model works:

$$\text{temp}_t = 10.92 + 0.62 \cdot \mathbf{1}[\text{recent}_t] + \varepsilon_t, \qquad \hat{\rho} = 0.195$$

The historical-window mean is 10.92degC and the estimated shift is 0.62degC, which means that the average summer minimum temperature in 1991 to 2020 is about 0.62 degC higher than in 1961 to 1990. The estimated $\hat{\rho}$ of 0.195 indicates mild year-to-year dependence in the residual deviations, so adjacent summers are somewhat related rather than fully independent. The 95% confidence interval [0.316, 0.924] lies entirely above zero, which means that the recent-window increase remains clearly positive even after the AR(1) correction has widened the uncertainty interval.

Table 1 and Figure 7 point in the same direction across almost all season-metric combinations. Every point estimate is positive, which indicates that the 1991 to 2020 window is warmer than 1961 to 1990 for 3 temperature types in every season. The clearest evidence appears in spring, where all three confidence intervals lie comfortably above zero and the estimated shifts are also the largest in magnitude. Summer and autumn show the same positive pattern, although with slightly smaller shifts than spring. Winter is the least certain case. Its point estimates are still positive throughout, but the lower bounds of all three intervals fall just below zero, so the evidence for a winter baseline shift is relatively weaker than other seasons.

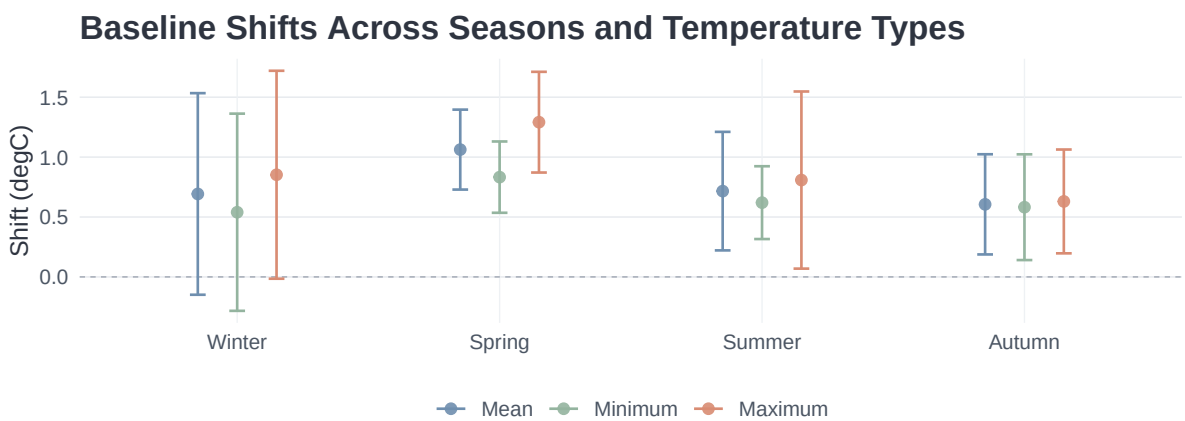

Figure 7: GLS AR(1) baseline shift estimates with 95% intervals by season and temperature type.

2.5.3 GLS AR(1): Long-Run Trend Model

The long-run trend model asks a different question from the baseline shift model. Instead of comparing two fixed periods, it asks how much temperature tends to change when year increases by one.

$$\text{temp}_t = \beta_0 + \beta_1 \cdot (t - \bar{t}) + \varepsilon_t, \qquad \varepsilon_t = \rho\,\varepsilon_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

Here, β_0 is the fitted temperature level at the average year of the series. β_1 is the annual slope, so $\beta_1 \times 10$ tells us how much temperature is estimated to increase per decade. The term $(t - \bar{t})$ means that the model measures each year relative to the middle of the observed time period, because this leaves the fitted slope unchanged, makes the intercept easier to interpret, and reduces collinearity between the intercept and slope. The definitions of ε_t, ρ, η_t, and σ^2 are otherwise unchanged from the shift model. Each season-metric combination is again fitted separately, now using the full 1879 to 2024 record, so the estimated slope is less sensitive to the exact placement of the two baseline windows and makes use of substantially more historical information. Spring maximum in Table 2 is used here as a worked example:

$$\text{temp}_t = 12.55 + 0.0114 \cdot (t - 1951.5) + \varepsilon_t, \qquad \hat{\rho} = 0.141$$

Table 2: GLS AR(1) long-run warming rates (degC per decade).

Season	$\beta_1 \times 10$: Mean	$\beta_1 \times 10$: Min	$\beta_1 \times 10$: Max
Winter	+0.097 [+0.040, +0.155]	+0.072 [+0.015, +0.129]	+0.123 [+0.065, +0.181]
Spring	+0.108 [+0.073, +0.143]	+0.102 [+0.072, +0.131]	+0.114 [+0.070, +0.159]
Summer	+0.089 [+0.056, +0.122]	+0.072 [+0.047, +0.096]	+0.106 [+0.060, +0.152]
Autumn	+0.125 [+0.094, +0.155]	+0.117 [+0.086, +0.148]	+0.132 [+0.099, +0.166]

$\beta_0 = 12.55$ is the fitted spring maximum temperature at the average year of the record, and $\beta_1 = 0.0114$ means that the fitted spring maximum temperature rises by about 0.0114 degrees Celsius when year increases by one. Multiplying that annual slope by ten gives 0.114 degrees Celsius per decade, which is the value reported in Table 2. The estimated $\hat{\rho}$ of 0.141 is clearly positive, so year-to-year dependence is not negligible in this series. The 95% confidence interval [0.07, 0.159] still remains entirely above zero, so the long-run warming signal remains positive after the AR(1) correction.

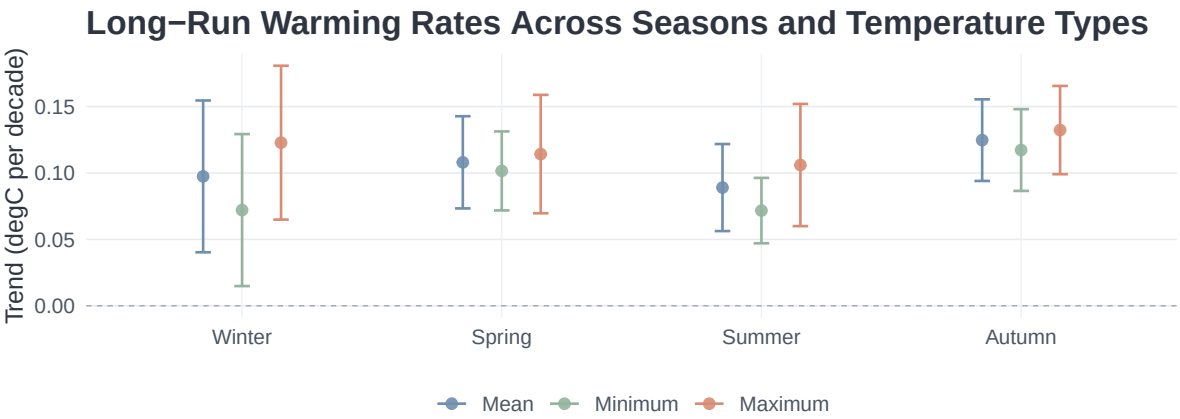

Figure 8: GLS AR(1) long-run warming rates with 95% intervals by season and temperature type.

Figure 8 confirms positive long-run warming across all four seasons and all three temperature metrics, with all 12 confidence intervals lying above zero. Autumn records the fastest mean warming rate at +0.125 degC per decade (95% CI $[+0.094, +0.155]$), while summer is the slowest at +0.089 ($[+0.056, +0.122]$). Worth highlighting is the winter result. Over the 60-year comparison window, the baseline shift was relatively weak because the confidence interval ($[-0.149, +1.534]$) straddles zero. Extending the analysis to the full 1879 to 2024 record resolves this, giving a clear positive trend of +0.097 degC per decade ($[+0.040, +0.155]$), which is a meaningful finding for a season that looked ambiguous in the earlier model.

Figure 6 helps explain why the long-run warming rates differ across seasons. A season tends to show a higher overall warming rate when its rolling 30-year temperature trend spends more time in positive territory. Autumn follows this pattern most clearly. Winter is the opposite case, with a deep and prolonged cooling phase through much of the mid-twentieth century that pulls down the long-run average and produces the weakest full-period warming rate of the four seasons. Spring and summer sit in between. This contrast also accounts for one of the more interesting findings in the results, spring records the largest recent baseline shift in Table 1 at +1.063 degC ($[+0.729, +1.396]$), yet this does not translate into the highest long-run warming rate, while autumn's smaller recent shift of +0.606 degC ($[+0.188, +1.024]$) comes with the strongest long-run average warming rate.

These intervals should be treated as uncertainty conditional on the current model. They do not account for model misspecification such as non-linearity in long-run warming, nor for structural variance change across the record, so they are best read as a floor on uncertainty rather than a complete picture.

2.5.4 Serial Dependence in Baseline and Long-Run Models

Figure 9 reports the estimated AR(1) coefficients, $\hat{\rho}$, for both the baseline shift model and the long-run trend model. These coefficients summarise how strongly residual deviations carry over from one year to the next after the mean structure has been fitted.

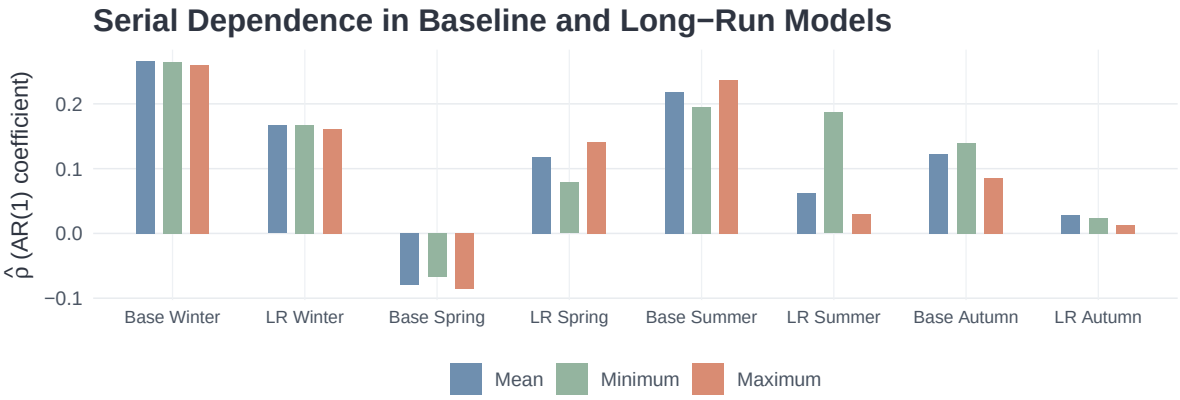

Figure 9: Estimated AR(1) serial dependence coeff from the baseline shift and LR trend models by season.

From Figure 9 we can see that serial dependence is not equally strong across the seasonal series. In the baseline shift model, winter and summer have the clearest positive AR(1) coefficients, with values mostly around 0.20 or above across the three temperature metrics, so year-to-year persistence is substantial enough to matter for inference. By contrast, baseline spring lies close to zero and is slightly negative, whilst long-run autumn is also very close to zero. Coefficients of that size should be interpreted cautiously, because they may reflect weak or unstable dependence rather than a statistically meaningful pattern. Autumn in the baseline model remains positive but is clearly weaker than winter and summer. The long-run trend model shows that dependence is still present after the trend has been fitted, especially in winter and in summer minimum, but the pattern is more mixed than in the baseline model. A notable feature is that long-run summer minimum remains clearly higher than the corresponding mean and maximum coefficients, which suggests that residual persistence is not distributed uniformly across the summer temperature range.

Overall, Figure 9 suggests that the AR(1) assumption is worth retaining as a practical working specification, because several series show clear positive persistence and ignoring it would risk understating uncertainty. At the same time, the near-zero cases show that dependence is not uniformly strong in every season, so AR(1) should be treated as an adequate overall correction rather than a perfect description of every individual series.

2.5.5 Model Comparison and Comparison Criteria

This section compares three dependence specifications under the same mean structure for each model family. OLS is included as the baseline specification where residuals are treated as independent, so it provides the simplest reference for fit and interval width. GLS AR(1) is included because the earlier dependence diagnostics reported non-negligible year-to-year persistence in several season-metric series. GLS AR(2) is included as a higher-order alternative to test whether a second lag materially improves fit once first-order dependence has already been modelled. For the baseline shift model, the mean structure remains the historical versus recent indicator. For the long-run trend model, the mean structure remains the centred linear year term.

The comparison records average AIC, average BIC, and average confidence-interval width for both model families. For the long-run trend family, average rolling one-step-ahead RMSE and its standardised form (NRMSE = RMSE/SD) are also reported because one-step forecasting is directly defined for a continuous time index and scale-normalised error supports cross-series comparison. Table 3 reports the baseline-shift family and Table 4 reports the long-run trend family.

Table 3: Model comparison: baseline-shift specification.

Model	Avg AIC	Avg BIC	Avg CI for β_1
OLS	156.562	162.845	0.913
AR(1)	157.095	165.472	1.022
AR(2)	158.186	168.657	0.927

Table 4: Model comparison: long-run trend specification.

Model	Avg AIC	Avg BIC	Avg CI for $\beta_1 \times 10$	Avg roll RMSE	Avg roll NRMSE
OLS	388.663	397.614	0.072	0.953	0.911
AR(1)	389.137	401.071	0.079	0.952	0.911
AR(2)	390.030	404.948	0.084	0.953	0.911

Since average AIC, average BIC, rolling one-step RMSE, and rolling NRMSE are all very close across OLS, AR(1), and AR(2), the metric differences are too small to identify a single dominant model. The deciding consideration is therefore the dependence evidence in Figure 9, where several series show residual persistence around 0.2, making the OLS independence assumption less credible for inference. AR(2) adds complexity but delivers little practical improvement over AR(1) on the reported metrics. For this reason, AR(1) is retained as the working specification, with OLS as a benchmark and AR(2) as a sensitivity alternative.

2.6 Validation and Robustness

The conclusions in Section 2.5.2 and Section 2.5.3 rely on the AR(1) model assumptions holding reasonably well and on the results not being driven by a small number of specific data points or modelling choices. This section checks both of those conditions directly, so that the reported confidence intervals and seasonal rankings can be taken at face value for decision use.

2.6.1 Residual Diagnostics

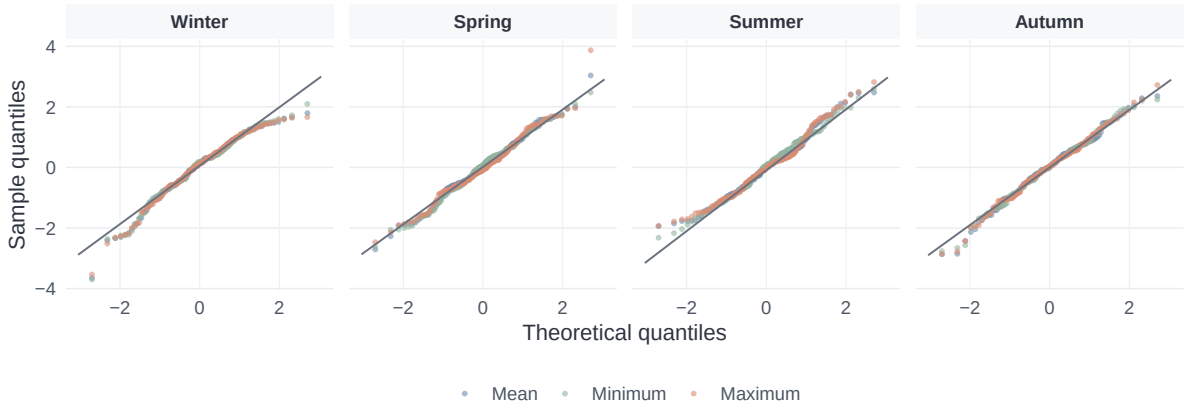

Figure 10: Q-Q plots of GLS AR(1) normalised trend residuals by season and temperature type.

Figure 10 checks whether the model residuals follow the normal distribution closely enough to trust the reported 95% confidence intervals. Across all four seasons, residuals track the reference line well through the central quantiles, which is where most of the estimation information lies, so the intervals can be taken at face value for the bulk of the distribution. The one area requiring care is the lower winter tail, consistent with the extreme winter years flagged in Section 2.3.3, spring and summer show mild upper-tail deviation but nothing that points to a systematic problem. In short, the normality assumption holds well enough for practical inference, with the caveat that winter tail-region estimates should be interpreted with slightly more caution.

2.6.2 Robustness Checks

Two robustness checks are applied to the long-run findings. The first uses a segmented AR(1) specification with a fixed 1975 breakpoint to test whether the warming conclusion holds once the model allows for a faster post-1975 slope. The second re-estimates the long-run trends after excluding the influential observations identified in Section 2.3.3 to confirm that the headline result is not driven by a small set of extreme years.

2.6.2.1 Segmented Linear Trend Sensitivity (Breakpoint at 1975) Figure 11 compares the baseline long-run AR(1) trend against a segmented AR(1) specification with a fixed breakpoint at 1975. The breakpoint is set with direct reference to Figure 6, which shows that warming rates begin to accelerate after around 1975 across all seasons. This check tests whether allowing a faster post-1975 slope changes directional conclusions or seasonal ranking.

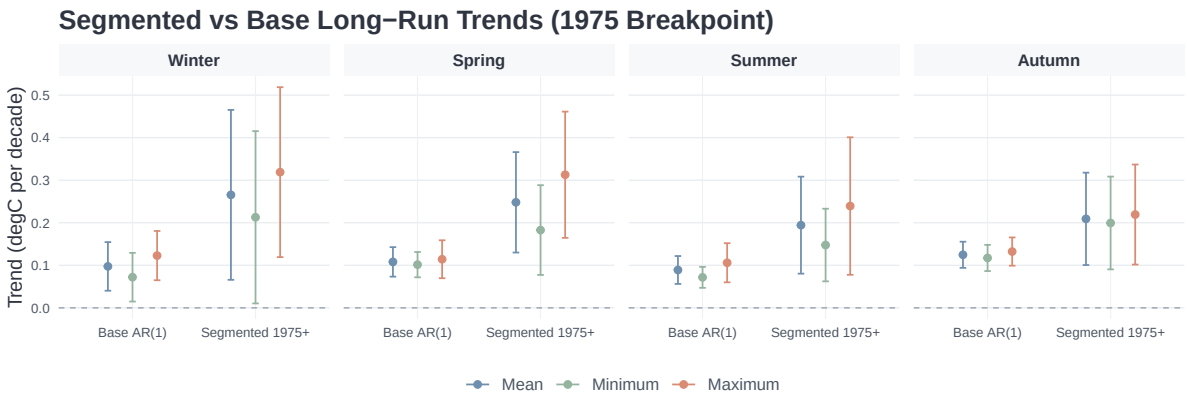

Figure 11: Segmented-trend sensitivity with a fixed 1975 breakpoint by season and temperature type.

Figure 11 shows a consistent pattern across all four seasons and all three temperature metrics, where the post-1975 segmented slopes are higher than the corresponding base AR(1) full-period slopes. The direction remains positive throughout, so the core warming conclusion is unchanged. The uplift is strongest in winter and spring, with particularly large increases in maximum temperature, whilst summer and autumn also rise in the same direction. This means the seasonal ranking of warming pace is not fixed across specifications. Full-period slopes are more balanced, whereas post-1975 slopes are more strongly led by winter and spring. Confidence intervals are wider under the segmented specification because the model estimates extra structure from a shorter effective period, yet the displayed intervals remain above zero. Therefore, the warming direction is robust and the warming rate accelerates after 1975.

2.6.2.2 Sensitivity to Influential Extreme Years Figure 12 evaluates sensitivity to influential points identified in Section 2.3.3. The excluded observations are winter 1963, summer 1976, autumn 2006, autumn 2022, and autumn 2023.

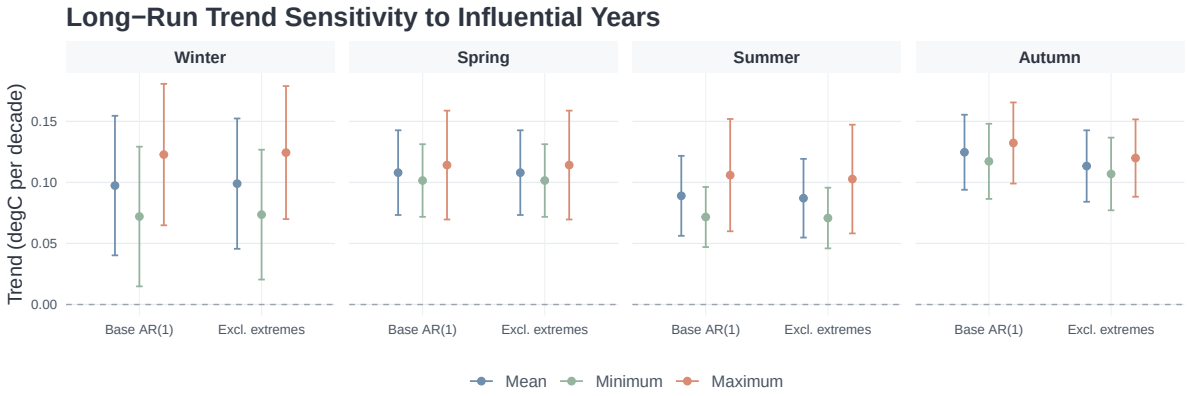

Figure 12: Sensitivity of long-run AR(1) trends to exclusion of flagged influential year-season observations.

Figure 12 indicates that excluding the flagged influential observations has only minor impact on the long-run trend estimates. Point estimates under the exclusion specification remain close to the base AR(1) estimates for every season-metric combination, and the confidence intervals overlap strongly. No directional reversal appears and the relative seasonal pattern remains stable. This indicates that the reported long-run warming trends are not driven by a small set of extreme years.

2.7 Monitoring and Material-Change Thresholds

Seasonal mean is used as the primary monitoring metric, and two baseline windows are used for monitoring. The 30-year window (1991–2020) is the primary decision baseline for all formal threshold calculations, anchored to the period when warming rates turned consistently positive across all four seasons (Figure 6). The 10-year window (2015–2024) serves as a short-horizon signal, rates in several seasons have moderated somewhat since around 2010. Comparing the two sets of thresholds shows directly whether the most recent decade has already shifted beyond the 30-year reference.

P10 and P90 are used as the reference band boundaries. Each marks the outer 10 per cent of the reference distribution, so a season falling below P10 or above P90 is a genuinely uncommon event relative to the 1991 to 2020 baseline, occurring about once in ten seasons under stable conditions. This frequency is low enough to carry a clear signal but common enough to remain sensitive to gradual change.

Figure 13 reports the P10 to P90 range for each season-metric combination from the 30-year and 10-year reference windows. A consistent upward shift in the 10-year band relative to the 30-year band would confirm that the most recent decade already sits above the primary reference distribution, making the 30-year P90 threshold increasingly likely to be exceeded in coming seasons. Therefore, a three-level status classification is used for planning purposes: - **Reference**: Oservation within P10 to P90. Continue routine monitoring. - **Watch**: Single-season value outside P10 or P90. Flag in briefing and track next season. - **Action**: Two consecutive seasonal breaches beyond P10 or P90, escalate with a seasonal impact note.

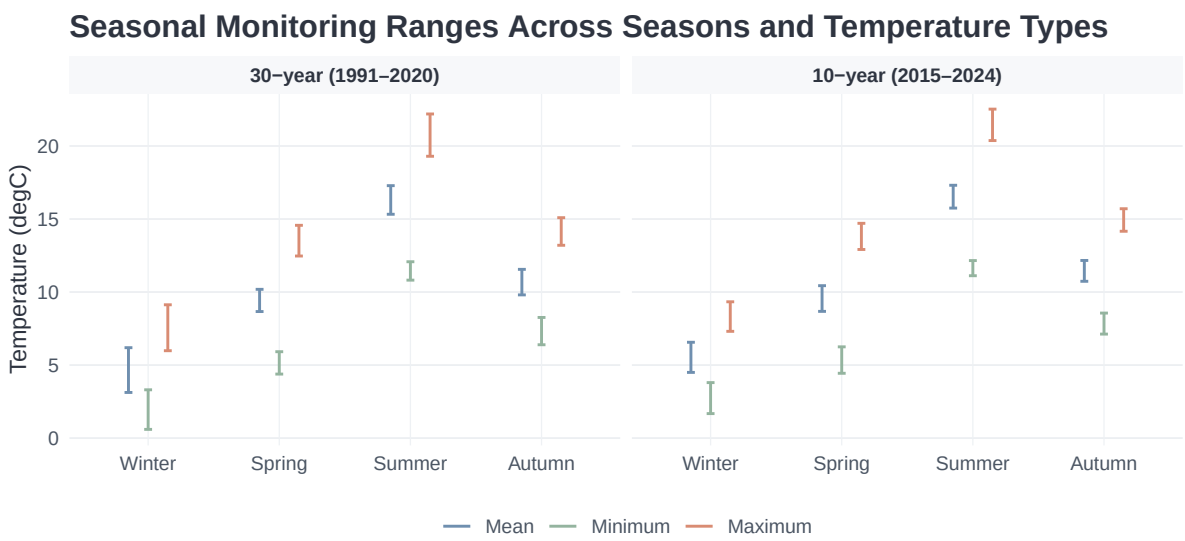

Figure 13: Seasonal P10 to P90 monitoring ranges from the 30-year and 10-year reference windows by season and temperature type.

2.8 Limitations, Validity, and Alternatives

2.8.1 Threats to Validity and Limitations

Internal Validity: Section 2.5.4 showed that $\hat{\rho}$ was generally around 0.1 to 0.2 across the fitted series. AR(1) therefore fits the climatic intuition that adjacent years are not fully independent, but it does not follow that the effect of ρ is large enough to materially change every inference. At the same time, AR(1) captures only one-step dependence and AR(2) extends this only slightly. Neither is sufficient to guarantee that more complex, longer-range, or non-linear residual structure has been fully represented.

External Validity: The conclusions apply to the HadCET Central England seasonal series and to annual seasonal temperature conditions at that regional scale. Their generalisability is limited in two ways. First, they do not directly establish finer spatial variation within the Midlands. Second, the dataset compresses temperature information into annual seasonal values and does not retain monthly resolution, so its temporal granularity is therefore limited, which makes it more suitable for season-level analysis than for finer-grained temperature variation.

Construct Validity: The monitoring framework in Section 2.7 is intentionally heuristic. The use of P10 and P90 as thresholds, together with the 30-year and 10-year reference windows, was chosen for interpretability and operational

usefulness in internal briefing rather than because these settings are the unique mathematical optimum. P10 and P90 are preferable to P25 and P75 for this purpose because they do not trigger as frequently, but they still represent an empirical compromise rather than a formally optimal cut-off. The same applies to the two reference windows. The 30-year window provides a longer baseline and the 10-year window provides a recent signal, yet both involve judgement rather than a single objectively correct choice.

2.8.2 Failure Mode That Would Change the Conclusion

The most plausible failure mode is that the model assumptions do not match the true error structure closely enough. Both Sections 2.5.2 and 2.5.3 assume that the innovation term η_t is approximately normal and use normal approximations to construct standard errors and 95% confidence intervals. If the true innovations are more skewed, more heavy-tailed, or more prone to extreme values than this assumption allows, then the interval estimates may misstate uncertainty. Section 2.6.1 suggests that the central residual distributions are broadly acceptable in most season-metric combinations, but the winter tail still departs visibly from the reference line. This means that the direction of warming may remain credible whilst the precision of some interval statements, especially in winter, is less secure than the model formally implies.

2.8.3 Alternative Approaches and Conditions for Preference

A plausible alternative is a dynamic linear model, which allows both the underlying trend and the error structure to evolve through time instead of imposing one fixed slope and one fixed dependence pattern over the full record. Relative to the GLS long run trend model in Section 2.5.3, this approach is better suited to the changing warming rates shown in Figure 6, and it would allow changes in the mean structure and residual dependence to be handled within one time-series framework.

It was not chosen here because the report required a common set of results that could be compared directly across seasons and temperature metrics, and the baseline-shift model together with the single long run slope provided a clearer decision-ready summary for that purpose. A dynamic linear model would become preferable if the main objective shifted from estimating recent contrasts and full-period average change to tracking how the pace of warming and the dependence structure evolve through time.

2.9 Traceability and Reproducibility

2.9.1 Claim-to-Evidence Map for Decision 1, 2, and 3

Claim 1. Recent decades are warmer than the earlier baseline in most seasons. (Table 1, Figure 7, Section 2.5.2)

Claim 2. Long-run seasonal warming is positive across all four seasons and across mean, minimum, and maximum temperature (Table 2, Figure 8, Section 2.5.3_

Claim 3. The warming pattern is sustained over the long run and is not adequately described as a single one-off shift (Figure 5, Figure 6, Figure 11, Sections 2.4.2, 2.4.3, and 2.6.2.1)

Recommendation 1. Internal seasonal briefing should use a threshold-based monitoring framework built from recent seasonal reference distributions (Figure 6, Figure 8, and Figure 11, Section 2.7)

2.9.2 Simplifications Planned for the Client-Facing Report

The client-facing report simplifies the technical material in three main ways. First, it translates model specifications and diagnostics into plain statements about change, consistency, and decision relevance rather than presenting equations or dependence parameters. Second, it reports a small number of headline seasonal comparisons and monitoring rules instead of the full set of technical estimates and robustness outputs. Third, it presents uncertainty in operational terms by explaining what should be treated with more caution and what would justify follow-up, rather than emphasising the statistical details of model checking. These simplifications are acceptable because they do not alter the direction of the findings, the relative ranking of seasons, or the practical recommendations that follow from the evidence. They reduce technical detail, but they do not remove any information that would materially change the client decision.

2.9.3 Repository Structure, Run Sequence, Environment, and Determinism

Github Repository: https://github.com/Yueran-Wang-Warwick/ST422-personal-portfolio-Yueran-Wang.git

`README.md` (run order), `.Rprofile/reports/.Rprofile` (project library activation), and `renv.lock` (dependency versions) ensure the reproducibility.

The execution sequence is fixed: start in the root directory, run `raw/data-cleansing.Rmd` to rebuild `processed/`, then render `reports/reports.Rmd`. Data are read from local `raw/` HadCET files, environment management uses `renv` (R 4.4.1). Determinism is high because the analysis pipeline does not use random sampling/simulation, which reruns the same outputs for every run.

2.9.4 Output Inventory and Remaining Reproducibility Risks

Outputs are saved with stable names in fixed folders: figures in `outputs/figures/`, tables in `outputs/tables/`, and the compiled report at `reports/reports.pdf`. Always render from project root, run `renv::restore()` before rerun on a new machine/session, and clear `cache/` plus `reports/cache/` when outputs appear stale or after dependency updates. These controls are sufficient for full audit replay under the current portfolio scope.

2.10 Appendix: Individual reflection

2.10.1 Key judgement calls you made

My main analytical judgement was to combine a recent baseline comparison with a long run trend analysis. I chose this structure because the client needed both a direct comparison with an older benchmark and a defensible account of whether the observed change formed part of a longer pattern. A more complex model might have described the process in greater detail, but it would have been harder to explain and defend in a consultancy context.

My main communication judgement was to organise the client report around four findings, each linked to one figure and one practical implication. I chose that structure because the brief was decision focused. It required clear conclusions about benchmarks, seasonal priorities, and monitoring rather than a full account of every analytical step.

2.10.2 Trade-offs

One trade-off arose from the data structure itself. The mean series extends further back than the minimum and maximum series, so I restricted the cross-metric analysis to their shared period from 1879 to 2024. This meant giving up some additional long run context from the mean series, but it avoided making comparisons across unequal historical windows.

I also used an AR(1) specification as the main working model and presented the monitoring thresholds as practical decision rules. This improved interpretability and kept the report auditable, but it also meant accepting a simpler dependence structure than the climate process may fully support. I managed that risk by reporting diagnostics, comparing alternative models, and checking that the main conclusions were stable.

2.10.3 What you learned and what you would do differently

This project taught me that consultancy quality depends on communication as much as on technical correctness. The most useful results were not the most detailed ones. They were the results that could be linked directly to a planning decision and defended under challenge. I also learned that uncertainty has to be translated into decision language rather than left in technical terms.

If I were doing the project again, I would separate the client and technical narratives earlier in the drafting process and test the monitoring section sooner with non-technical readers. That part of the report required the most revision, which showed me that clarity and defensibility need to be built into the analysis from the start.

2.11 Appendix: Evidence of group contact and professional working

Contact log: `2026-02-17_meeting_APPROVED.md`

Minutes of Meeting

Project: Client Brief 6
Group: Group 3
Meeting type: Weekly sync
Date: 2026/02/17
Time: 15:00-17:00
Location/Platform: FAB collection space
Minute-taker: Shan Xue

1. Attendees

- Alex Zheng (present)
- Lewis Parry (present)
- Yueran Wang (present)
- Yunting Chen (present)
- Priya Kumar (present)
- Shan Xue (present)

2. Agenda

1. Confirm deliverables required and agree on client brief selection.

2. Assigning work for each individual this week.

3. Find questions for clients.

4. Decide to use GitHub for issue board/task track.

3. Key Decisions Made

Decision: Assign weekly responsibilities to individual team members.
Rationale: To ensure accountability and timely completion of Deliverables 1-4 before Week 7 Monday 0900.
Implication: Clear ownership established for each required output:
- Alex - upload `SELECTION.md`
- Yueran - upload scope note and allocate roles
- Lewis - complete the management plan
- Priya - create initial `README.md`
- Yunting - maintain action log and approved minutes
- Shan - draft meeting minutes

4. Discussion notes

1. Confirmed required outputs for Week 7: SELECTION.md, management plan (3 formats), updated shared space, repository structure and README.

2. Decided to create branch per issue, then ask others to review the code after pushing.

5. Action log

Action	Owner	Deadline	Status	Evidence Pointer
Create repo structure and setup contact log	Lewis Parry	2026-02-17	Completed	Repository initialised (main branch)
Sync raw data to repository and upload files	Yueran Wang	2026-02-17	Completed	commit dd938c6e082c6a9ba7c2bc03056412beaf0c695c
Create GitHub issue board	Lewis Parry	2026-02-20	Completed	GitHub Issues
Upload draft minutes	Shan Xue	2026-02-17	Completed	`Minutes/2026-02-17_meeting_DRAFT.md`
Complete action log and approved minutes	Yunting Chen	2026-02-20	Completed	`Minutes/2026-02-17_meeting_APPROVED.md`
Upload `SELECTION.md`	Alex Zheng	2026-02-20	Completed	`SELECTION.md`
Upload `Scope_Notes.md` and `Role_Allocation.md`	Yueran Wang	2026-02-20	Completed	`Scope_Notes.md`, `Role_Allocation.md`
Upload management plan (`.md`, `.html`, `.pdf`)	Lewis Parry	2026-02-20	Completed	`management_plan.md`
Create initial `README.md`	Priya Kumar	2026-02-20	Completed	`README.md`

6. Risks / blockers / dependencies

- None identified at this stage.

7. Reproducibility / QA checks

1. Agreed branch-per-issue workflow.

2. All merges require peer review.

8. Questions for Client

1. Are there specific temperature metrics that should be prioritised?

2. In practical decision terms, what outcome would you consider most useful from this analysis?

3. Do you have a preferred historical baseline period for comparison, or should we propose one based on statistical stability?

9. Next meeting

Date/time: Week 7 Monday
Location/Platform: In person
Provisional agenda: 1. Finalise questions for client.
2. Update action log and assign Week 7 tasks. —

Contact log: `2026-02-23_meeting_APPROVED.md`

Minutes of Meeting

Project: Brief 6
Group: Group 3
Meeting type: Weekly sync
Date: 2026/02/23
Time: 10:37-10:50, 11:30-11:55
Location/Platform: MB0.07
Minute-taker: Shan Xue

1. Attendees

- Alex Zheng (present)
- Lewis Parry (present)
- Yueran Wang (present)
- Yunting Chen (present)
- Priya Kumar (present)
- Shan Xue (present)

2. Agenda

1. Client requested meeting: interim progress.

2. QA discussion with client.

3. Action log evidence pointer structure

3. Key Decisions Made

- **Decision 1:** The team will independently determine the most appropriate temperature metrics for modelling and justification.
  Rationale: The client has no preferred metric and expects evidence-based justification.
  Implication: Metric selection must be defensible, clearly documented, and supported by analysis.

- **Decision 2:** The team will propose a statistically stable historical baseline period.
  Rationale: No preferred baseline was provided by the client.
  Implication: Baseline sensitivity analysis will be included in robustness checks.

- **Decision 3:** Action log updates must include GitHub evidence links in the "Evidence Pointer" column after peer review.
  Rationale: To strengthen auditability and reproducibility.
  Implication: All completed actions must reference an Issue, PR, or commit link.

4. Discussion notes

Q1. Are there specific temperature metric that should be prioritised?
A: Client has no preference; we should choose appropriate metrics ourselves and justify with evidence.

Q2: In practical decision terms, what outcome would be most useful from this analysis?
A: Most useful outcome aligns with the first point in the client brief; outputs must be robust and defensible. Client is relying on our analysis and expects it to stand up to scrutiny and answer likely follow-up questions.

Q3: Do you have a preferred historical baseline period for comparison, or should we propose one based on statistical stability?
A: No preferred historical baseline period.

Q4. Action log evidence pointer structure
A: After peer review, action log should be updated in the approved minute and the related GitHub link should be attached in the "pointer period" section.

5. Action log

Action	Owner	Deadline	Status	Evidence Pointer
Data cleaning and validation	Alex Zheng	2026-02-25	Completed	`data/processed/data_max_cleaned.csv`, `data/processed/data_mean_cleaned.csv`, `data/processed/data_min_cleaned.csv`
EDA visualisation and narrative integration	Yueran Wang	2026-02-25	Completed	`reports.Rmd`
Upload draft minutes	Shan Xue	2026-02-23	Completed	`Minutes/2026-02-23_meeting_DRAFT.md`
Complete action log and approved minutes	Yunting Chen	2026-02-27	Completed	`Minutes/2026-02-23_meeting_APPROVED.md`
Peer review of data cleaning	Lewis Parry	2026-02-27	Completed	PR #6 reviewed and approved
Peer review of EDA outputs	Yunting Chen	2026-02-27	Completed	PR #5 reviewed and approved
Peer review of README file	Yunting Chen	2026-02-27	Completed	PR #7 reviewed and approved
Peer review of Management plan	Alex Zheng	2026-02-27	Completed	`management_plan.md`
Draft section titles and report structure	Lewis Parry	2026-02-27	Completed	Report outline draft
Confidence statement (Client Q1)	Yueran Wang	2026-02-28	Completed	`client_meeting_Q1_brief6.md`
Uncertainty and Bias Framing (Client Q2)	Shan Xue	2026-02-28	Completed	`client_meeting_q2_uncertainty.md`
Robustness checks (Client Q3)	Priya Kumar	2026-02-28	Completed	——-

6. Risks / blockers / dependencies

- **Dependency:** Completion of peer review before merging modelling outputs or file change.
  Due: 2026-02-27
  Status: Open

7. Reproducibility / QA checks

- ☐ Figures/tables regenerate without manual edits

- ☐ Peer review completed (link evidence)

8. Next meeting

Date/time: Week 8 Monday
Location/Platform: MB0.07
Provisional agenda:
1. Present modelling results and robustness checks.
2. Confirm baseline selection rationale.
3. Review action log and QA evidence. —

Contact log: `2026-03-04_meeting_APPROVED.md`

Minutes of Meeting

Project: Brief 6
Group: Group 3
Meeting type: Weekly sync
Date: 2026/03/04
Time: 13:30-15:00
Location/Platform: WA1.01
Minute-taker: Shan Xue

1. Attendees

- Alex Zheng (present)
- Lewis Parry (present)
- Yueran Wang (apologies)
- Yunting Chen (present)
- Priya Kumar (present)
- Shan Xue (present)

2. Agenda

1. Review questions form the client meeting on Monday (2026-03-02).

2. Assign the report for this week.

3. Review the current progress of the project and the code developed so far.

3. Key Decisions Made

- **Decision:** Search for additional datasets or government sources related to the energy question mentioned in the client meeting.
  Rationale: Our current data cannot support answering the energy question. With the client's permission, we will carry out some additional research.
  Implication: We will spend some time this week, and possibly next week, searching for evidence that could support our conclusions.

4. Discussion notes

Q1. Why is the temperature still increasing even though the UK is using cleaner energy, and should the UK return to older energy sources?
A: Global cumulative emissions are still increasing. Emissions depend on both energy demand and the carbon intensity of energy, and many countries still rely heavily on fossil fuels. Climate inertia also means that CO2 stays in the atmosphere for a long time and oceans absorb much of the excess heat, which can delay temperature changes. Therefore, returning to older energy sources would not solve the problem.

Q2. What statistical methods are used in our analysis?
A: The client asked about the use of GLM. We agreed that the report should briefly explain the method so that it is easier for the client to understand.

Q3. Can we use additional data to strengthen our conclusions and suggestions?
A: Yes. The team agreed to search for additional datasets or external sources that could help support our conclusions.

5. Action log

Action	Owner	Deadline	Status	Evidence Pointer
Investigate the energy question raised by the client	Alex Zheng	2026-03-06	Completed	`2026-03-02_client_briefing_meeting.docx`
Upload draft minutes	Shan Xue	2026-03-04	Completed	`Minutes/2026-03-04_meeting_DRAFT.md`
Complete action log and approved minutes	Yunting Chen	2026-03-08	Completed	`Minutes/2026-03-04_meeting_APPROVED.md`
Complete next week client status report	Lewis Parry	2026-03-06	Completed	`Week9_client_status_report_part2.md`
Complete next week client status report	Shan Xue	2026-03-06	Completed	`Week9_client_status_report_part1.md`
Commenting code and clean run	Priya Kumar	2026-03-16	In progress	——-
Robustness sensitivity check	Yunting Chen	2026-03-16	In progress	——-
Analysis of effect of clean energy	Yueran Wang	2026-03-06	In progress	——-

6. Risks / blockers / dependencies

- **Risk:** The client may misinterpret the rise in temperature as evidence that clean energy is not effective and consider returning to older energy sources.
  Due: Next client meeting
  Status: Open
- **Risk:** Seasonal results may be influenced by a small number of unusually hot or cold years.
  Due: After robustness checks
  Status: Open

7. Reproducibility / QA checks

☐ Figures/tables regenerate without manual edits

☐ Peer review completed (link evidence)

8. Next meeting

Date/time: Week 9 Monday
Location/Platform: MB0.07
Provisional agenda:

—

Contact log: `2026-03-09_meeting_APPROVED.md`

Minutes of Meeting

Project: Client Brief 6
Group: Team 3 **Meeting type:** Weekly sync **Date:** 2026-03-11 **Time:** 13:30-14:57 **Location/Platform:** WA1.15A
Minute-taker: Shan

1. Attendees

- Yunting Chen (present)

- Yueran Wang (present)
- Alex Zheng (present)
- Shan Xue (present)
- Lewis Parry (present)
- Priya Kumar (present)

2. Agenda

1. Go through notes from client meeting on Monday
2. Split the work for week 10 client report
3. Discussion of 'lack of work to distribute.'

3. Key decisions made

- **Decision 1:** Defer the clean energy policy question for now
  Rationale: This question is outside the scope of the current brief and was not requested by the client at this stage

  Implication: We will address this topic in the presentation, if needed, with appropriate supporting evidence and clear scope boundaries.

- **Decision 2:** Assign at least two team members to each critical task

  Rationale: Following discussion with the lecturer, pairing team members on key tasks will improve continuity, enable faster handovers, and reduce delivery risk if availability changes.

  Implication: Each task will have at least two members to ensure coverage and timely progress.

4. Action log

Action	Owner	Deadline	Status	Evidence Pointer
Code comment	Priya Kumar	15/03/2026	Not Started	
Update readme	Priya Kumar	15/03/2026	Not Started	
Robustness check	Yunting Chen	15/03/2026	Not Started	
Modelling	Yueran Wang	15/03/2026	Completed	
Week 10 client report	Shan Xue	13/03/2026	Completed	
Week 10 client report	Alex Zheng	13/03/2026	Not Started	
Week 10 client report	Lewis Parry	13/03/2026	Not Started	
Convert client report into prose	Lewis Parry	13/03/2026	Not Started	

5. Reproducibility / QA checks

- ☐ Code comment
- ☐ Week 10 client report checking

File name suggestion: `minutes_YYYY-MM-DD.md`

Contact log: `2026-03-11_meeting_APPROVED.md`

Minutes of Meeting

Project: Brief 6
Group: Group 3
Meeting type: Weekly sync
Date: 2026/03/11
Time: 13:30-15:00
Location/Platform: WA1.15A
Minute-taker: Shan Xue

1. Attendees

- Alex Zheng (present)

- Lewis Parry (apologies)
- Yueran Wang (present)
- Yunting Chen (present)
- Priya Kumar (present)
- Shan Xue (present)

2. Agenda

1. Review questions form the client meeting on Monday (2026-03-09).

2. Assign the report for this week.

3. Key Decisions Made

- **Decision 1:** Defer the clean energy policy question for now.
  Rationale: This question is outside the scope of the current brief and was not requested by the client at this stage.
  Implication: We will address this topic in the presentation, if needed, with appropriate supporting evidence and clear scope boundaries.

- **Decision 2:** Assign at least two team members to each critical task.
  Rationale: Following discussion with the lecturer, pairing team members on key tasks will improve continuity, enable faster handovers, and reduce delivery risk if availability changes.
  Implication: Each task will have at least two members to ensure coverage and timely progress.

4. Discussion notes

- When writing for the client, the report should focus more on interpretation and conclusions, with less emphasis on technical or programming details.

- The report should maintain a simple and consistent style suitable for non-expert readers, including clear plots and consistent formatting.

- The robustness checks completed so far were viewed positively, but explanations should remain non-technical and easy to understand.

5. Action log

Action	Owner	Deadline	Status	Evidence Pointer
Convert client report into prose	Lewis Parry	2026-03-16	Completed	——-
Complete next week client report	Lewis Parry	2026-03-13	Completed	`ST422_Week10_Client_Update_part2.m`
Complete next week client report	Alex Zheng	2026-03-13	Completed	`ST422_Week10_Client_Update_part1.m`
Complete next week client report	Shan Xue	2026-03-13	Completed	`ST422_Week10_Client_Update_part3.m`
Commenting code and clean run	Priya Kumar	2026-03-16	Completed	`reports.rmd`
Update readme file	Priya Kumar	2026-03-16	Completed	`README.md`
Robustness sensitivity check	Yunting Chen	2026-03-16	Completed	`reports.rmd`
Main modelling	Yueran Wang	2026-03-16	Completed	`reports.rmd`

6. Risks / blockers / dependencies

- **Risk:** The client may misinterpret the rise in temperature as evidence that clean energy is not effective and consider returning to older energy sources.
  Due: Next client meeting
  Status: Open

- **Risk:** Seasonal results may be influenced by a small number of unusually hot or cold years.
  Due: After robustness checks
  Status: Open

7. Reproducibility / QA checks

☐ Figures/tables regenerate without manual edits

☐ Peer review completed (link evidence)

8. Next meeting

Date/time: Week 10 Monday
Location/Platform: MB0.07
Provisional agenda:

- `appendix/group_contact_log/group_contact_log_summary.md` contains the consolidated group contact log with date, attendees, duration, agenda, actions, owners, deadlines, status, and evidence pointers. This is a summary only. Full meeting files are in `appendix/group_contact_log/`.
- `appendix/audit_reports/individual_action_log.md` contains my individual actions undertaken and completed with task, dates, outputs, evidence links, and outcomes.

Both summaries are compiled from approved meeting minutes in `appendix/group_contact_log/` and the audit export in `appendix/audit_reports/AUDIT_REPORT.html`.

2.12 Appendix: Management plan

Use this appendix to show **how you planned and managed delivery**, not just what analysis you produced. Keep it evidence-based and practical.

Workplan and milestones

Project Roadmap: Phases of Work

Phase 1: Triage

Clarify decision and scope

- **Purpose**: Agree on scope and limitations.
- **Key Activities**:
  - Read project brief.
  - Confirm dataset (HadCET).
  - Define excluded variables.
- **Deliverables**: Signed-off `SELECTION.md` and scope notes.
- **Owner**: **Shan Xue** (Project Lead)
- **Deadline**: Week 6
- **Definition of Done**: Scope is formally agreed upon by all leads.

Phase 2: Data Readiness

Ensure usable dataset

- **Purpose**: Acquire and validate data.
- **Key Activities**:
  - Download HadCET v2.1.0.0.
  - Check for missing values.
  - Format dates and schema.
- **Deliverables**: Clean `.csv` file and data dictionary.
- **Owner**: **Alex Zheng** (Data Steward)
- **Deadline**: Week 7
- **Definition of Done**: Reproducible cleaning script runs without error.

Phase 3: Analysis

Produce evidence

- **Purpose**: Generate statistical findings.

- **Key Activities**:
  - Calculate 30-year baselines.
  - Run trend tests (e.g., Mann-Kendall).
  - Plot time series data.
- **Deliverables**: Draft figures and summary tables.
- **Owner**: **Yueran Wang** (Analysis Lead)
- **Deadline**: Week 8
- **Definition of Done**: All claims are supported by p-values or confidence intervals.

Phase 4: Interpretation & Reporting

Write client output

- **Purpose**: Translate statistics into business meaning.
- **Key Activities**:
  - Draft internal briefing.
  - Interpret trends vs. step-changes.
- **Deliverables**: Draft report (RMarkdown).
- **Owner**: **Lewis Parry** (Reporting Lead)
- **Deadline**: Week 9
- **Definition of Done**: Narrative aligns strictly with evidence; no over-claiming.

Phase 5: QA

Make it defensible

- **Purpose**: Verify correctness and reproducibility.
- **Key Activities**:
  - Conduct peer code review.
  - Verify figure labels and units.
  - Re-run full pipeline on a fresh machine.
- **Deliverables**: Final `report.html` and QA log.
- **Owner**: **Priya Kumar & Yunting Chen** (QA Leads)
- **Deadline**: Week 10
- **Definition of Done**: A fresh run reproduces identical outputs to the draft.

Milestones and deadlines

Milestone	What it means	Why it matters	Due date (Est.)	Owner(s)	Status
Scope agreed	Brief selected, out-of-scope defined	Prevents scope creep	Week 6	Shan Xue	**Done**
Data freeze	Clean dataset version tagged	Stops moving targets	Week 7 (Mon)	Alex Zheng	**Done**
First figures	Core plots/tables drafted	Enables narrative early	Week 8	Yueran Wang	**Done**
Draft Report	Full text draft ready	Reviews can begin	Week 9	Lewis Parry	**Done**
Code Review	Code checked for logic/style	Ensures correctness	Week 9	Priya/Yunting	**Done**
Final QA sign-off	Checklist complete	Confident submission	Week 10	Priya/Yunting	**Done**

Allocation of responsibilities

Role	Responsibilities	Outputs owned	Primary	Backup
Project Lead	Keep plan on track; chair meetings; remove blockers	Minutes and task board	**Shan Xue**	Lewis Parry

Role	Responsibilities	Outputs owned	Primary	Backup
Data Steward	Data cleaning; data dictionary; versioning	Clean dataset and scripts	**Alex Zheng**	Yueran Wang
Analysis Lead	Main modelling/summaries; method notes	Results tables/figures	**Yueran Wang**	Alex Zheng
Reporting Lead	Client report and exec summary	Drafts and final text	**Lewis Parry**	Shan Xue
QA Lead	Check claims, figures, reproducibility	QA checklist and log	**Priya Kumar, Yunting Chen**	Alex Zheng

Risk register

Risk (what could go wrong)	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation (what you will do)	Owner	Status
Scope creep	M	H	Freeze scope in `SELECTION.md`; record out-of-scope items.	Shan Xue	Open
Data interpretation errors	L	H	Peer review of statistical claims (check p-values/CIs).	Priya/Yunting	Open
Git merge conflicts	M	M	Pull before push; communicate before editing same file.	Alex Zheng	Open
Late integration	H	H	Set "First Figures" milestone early (Week 8).	Lewis Parry	Open
Reproducibility failure	L	H	Run full pipeline on a fresh clone weekly.	Priya/Yunting	Open

Quality assurance plan

Data quality checks (Owner: Alex Zheng)

Check	How it is performed	Pass/Fail rule	When run
Missingness	Scripted summary	Flag if key months are missing in recent decades	Each update
Duplicate check	ID check on Year/Month	No duplicate time points	Each update
Range check	Rule-based	Temps must be reasonable (e.g., -20°C to +40°C)	Each update

Analysis robustness checks (Owner: Yueran Wang)

Robustness check	Why it matters	What you change
Baseline Period	Results depend on reference	Compare 1961-1990 w/ 1981-2010
Trend Method	Parametric assumptions	Compare Linear Regression w/ Mann-Kendall

Robustness check	Why it matters	What you change
Data Source	Measurement artifacts	Check Min/Max trends vs Mean trends

Review process (peer review, code review, figure checking)

Suggested approach:

- Schedule review as a milestone (not a last-minute activity).
- Use a simple checklist so reviewers know what "good" looks like.

Review process (peer review, code review, figure checking)

Review type	What is reviewed	Reviewer(s)	Checklist / criteria used	How feedback is recorded	Date	Evidence link
Code review	Reproducibility, style, logic	**Shan Xue**	; 1. No hard-coded paths; 2. MK-test logic correct	GitHub Pull Request (PR) comments	03 Mar 2026	https://github.com/Yueran-Wang-Warwick/ST422-Team-3/pull/26#pullrequestreview-4058872496

Reproducibility check (Owner: Priya Kumar, Yunting Chen)

- **Goal:** Run the full pipeline on a "clean" machine/account.
- **Command:** `rmarkdown::render("report/report.Rmd")`
- **Success Criteria:** Output HTML/PDF matches the submitted version exactly.